Wawes - L3

$x = A\cos(\omega t+\alpha) \rightarrow \frac{d^2x}{dt^2} = -\omega^2 x$

$y = A\sin(\omega t+\alpha) \Rightarrow \frac{d^2y}{dt^2} = -\omega^2 y$

$\Longleftrightarrow \Rightarrow \begin{cases} \frac{d^2z}{dt^2} = \frac{d^2}{dt^2}(x+iy) = \frac{d^2x}{dt^2} + i\frac{d^2y}{dt^2} \\ = -\omega^2(x+iy) \end{cases}$

$z = x+iy = A e^{[i(\omega t+\alpha)]}$

$\frac{dz}{dt} = \underbrace{A e^{[i(\cdot)]}}_{z} \underbrace{\frac{d}{dt}[\ ]}_{i\omega}$

$= i\omega z$

$\frac{d^2z}{dt^2} = \frac{d}{dt}\underbrace{\left(\frac{dz}{dt}\right)}_{i\omega z} = i\omega(i\omega z) = i^2\omega^2 z = -\omega^2 z$

$\frac{d^2x}{dt^2} + \omega^2 x = 0$ ($\ddot{x}$)

$x(t) = c e^{\lambda t}, \quad \ddot{x} = \lambda^2 c e^{\lambda t}$

$(\lambda^2+\omega^2) c e^{\lambda t} = 0 \Rightarrow$

$\lambda^2 = -\omega^2 \Rightarrow \lambda_{1,2} = \pm i\omega$

$x_1(t) = c_1 e^{i\omega t}, \quad x_2(t) = c_2 e^{-i\omega t}$

$x(t) = c_1 e^{i\omega t} + c_2 e^{-i\omega t}$

$e^{\pm i\omega t} = \cos(\omega t) \pm i\sin\omega t$

$x(t) = A\sin\omega t + B\cos\omega t$

A, B - reel

$x = \ldots$

B

A

c_1

c

$x = \ldots$

C

x

$\cos(\omega t) \pm i \sin \omega t$

$A \sin \omega t + B \cos \omega t$

- real

$$x = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$= A' \sin \omega t + B' \cos \omega t$$

$$= A \sin(\omega t + \varphi)$$

$$B' = C_1 + C_2$$

$$A' = i(C_1 - C_2)$$ | HW

$$C_1 = \frac{1}{2}(B' - iA')$$

$$C_2 = \frac{1}{2}(B' + iA') = C_1^*$$

$$x = C_1 e^{i\omega t} + C_1^* e^{-i\omega t}$$

$$C_1 = |C_1| e^{i\alpha} = C e^{i\alpha}$$

$$x = C e^{i(\omega t + \alpha)} + C e^{-i(\omega t + \alpha)}$$

$$= 2C \cos(\omega t + \alpha) = A \cos(\omega t + \alpha)$$

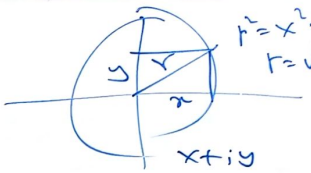
A_{New}

$$z = x + iy = r\cos\theta + i r \sin\theta$$

$$= r(\cos\theta + i \sin\theta)$$

$$= r e^{i\theta}$$

$$= |z| e^{i\theta}$$



$r^2 = x^2 + y^2$

$r = \sqrt{x^2 + y^2}$

$$x(t)\big|_{t=0} = x(0) = x_0$$

$$\frac{dx(t)}{dt}\Big|_{t=0} \equiv \dot{x}(0) \equiv \dot{x}_0 \equiv v_0$$

$$\equiv v(0)$$

$$x(t) = A \cos(\omega t + \alpha)$$

$$x(0) = x_0 = A \cos\alpha$$

$$\dot{x}(0) = v_0 = -\omega A \sin\alpha$$

$$A \sin\alpha = -v_0/\omega$$

$$A \cos\alpha = x_0$$

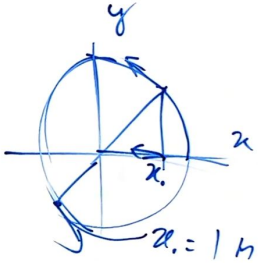
$$A = \left[x_0^2 + (v_0/\omega)^2\right]^{1/2} = \sqrt{2}$$

$$\tan\alpha = \left(-\frac{v_0}{\omega x_0}\right) = 1$$

$$\alpha = \frac{\pi}{4}, \cancel{\frac{5\pi}{4}}$$

$\alpha = \frac{\pi}{4}$

$$= \sqrt{2} \sin\left(t + \frac{\pi}{4}\right)$$



$x_0 = 1$ m

$v_0 = -1$ m/s

$\omega = 1$

24/6/26

Waves - L3

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = -\omega^2 x$$

$$mv\frac{dv}{dt} = -kx\cdot\frac{dx}{dt}$$

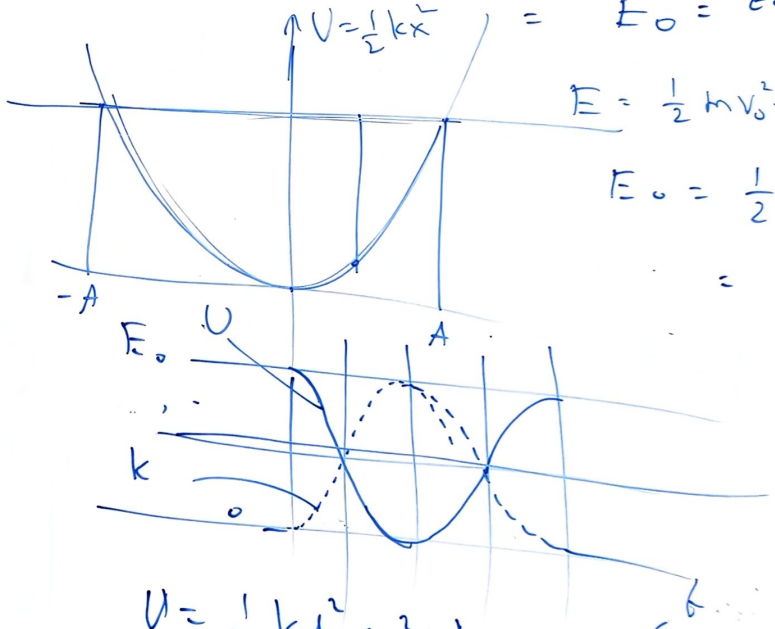
$$\frac{d}{dt}\left(m\frac{v^2}{2}\right) = -\frac{d}{dt}\left(\frac{k}{2}x^2\right)$$

$$\int_{v_0}^{v} d\left(m\frac{v^2}{2}\right) = -\int_{x_0}^{x} d\left(\frac{k}{2}x^2\right)$$

$$\frac{m}{2}(v^2 - v_0^2) = -\frac{k}{2}(x^2 - x_0^2)$$

$$\frac{1}{2}mv^2 + \frac{k}{2}x^2 = \frac{1}{2}mv_0^2 + \frac{k}{2}x_0^2$$

E_0 = total mechanical energy



$$E = \frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2$$

$$E_0 = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2 A^2$$

$$= \frac{1}{2}m(\omega A)^2$$

$$U = \frac{1}{2}kA^2\cos^2\omega t$$

$$K = \frac{1}{2}m\omega^2\sin^2\omega t$$